Iowa State University

Electrical and Computer Engineering

EE 2300

Design Project Report

Date of Experiment: April 9, 2026

Date of Report Submission: April 22, 2026

Name of Group Members: Michael Santiago, Preston Gaskill

Preston Gaskill

SEE SUBMISSION COMMENTS FOR A VIDEO DEMONSTATION

We were unable to demo during the week of April 13th. I emailed the TAs this video last Monday, but I haven't received a response. However, I was told during lab that if we attached a video to our final report, our demo points could be graded based on it.

Introduction

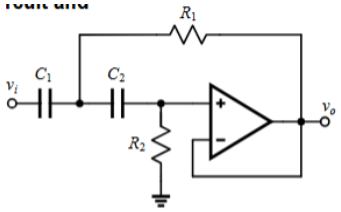
In our project, we implemented a frequency-rate pulser circuit that responds to an audio signal. The system is designed to take a single audio signal and separate it into three distinct frequency ranges: low, middle, and high, where each corresponds to a different LED color and turns on accordingly when active. We get to determine the values of these ranges. The overall design must operate with no more than two DC power supplies.

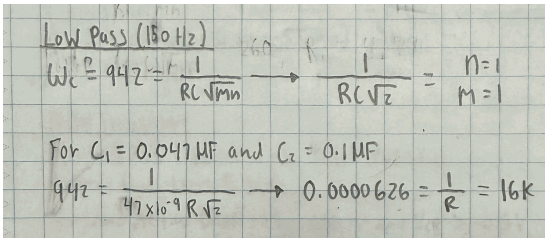
Description of the Design and Design Process

For our design, we knew we would need an amplifier for the audio signal, so when choosing between active and passive filters, we created three active second-order filters: a lowpass (Sallen-key), a bandpass (Multiple Feedback Filter), and a highpass (KRC Filter High), each passing its respective frequency band. We then ensured that our filter designs matched the general filter equation forms from the in-class slides and set two Q values: one for the high-pass and one for the bandpass. Our equations yielded low-error resistor values, but due to component availability and circuit simplicity we substituted certain resistor values for components found in our kit. All of our substitutions fell within a 10% range of the original design, which resulted in similar behavior, but still differed slightly

Low-pass

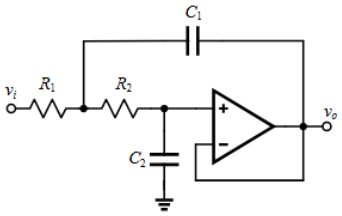
$$H(s) = G_0 \frac{\omega_c^2}{s^2 + \left(\frac{\omega_c}{Q}\right)s + \omega_c^2}$$

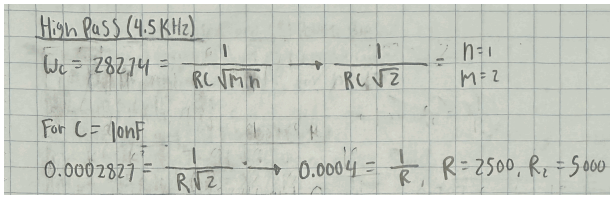




High-pass

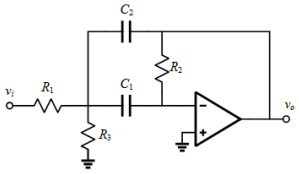
$$H(s) = G_\infty \frac{s^2}{s^2 + \left(\frac{\omega_c}{Q}\right)s + \omega_c^2}$$

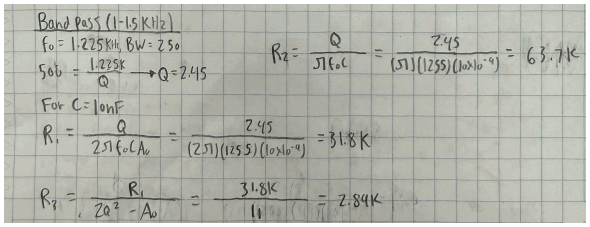




Band-Pass

$$H(s) = G_0 \frac{\left(\frac{\omega_0}{Q}\right)s}{s^2 + \left(\frac{\omega_0}{Q}\right)s + \omega_0^2}$$

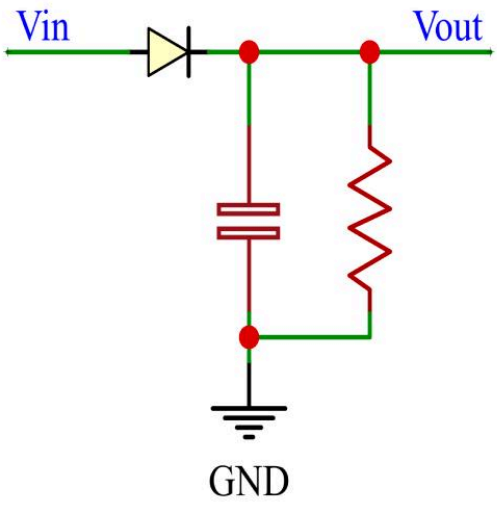




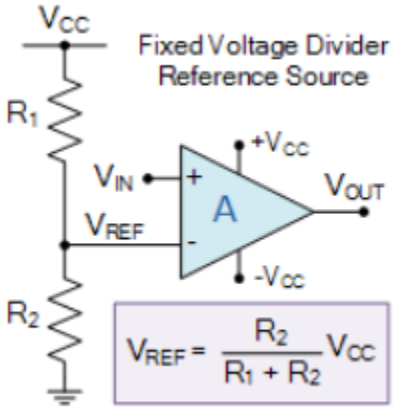
From here, we set our cutoff frequencies to about 115 Hz, 900-1150 Hz, and 4500 Hz for each of our respective filters, determined the ideal calculated values, and matched them to our available components.

We built all our circuits in LTSpice, but after a lot of experimentation, we found that there were certain conditions that LTSpice could not account for, so ultimately, for any physical circuit problems we had, it was better to use LTSpice as a reference (not a guide) and troubleshoot them in person. So with this and the component changes from earlier, our filter cutoff frequencies ended up being more like 175Hz, 800Hz - 2000Hz, and 4000Hz.

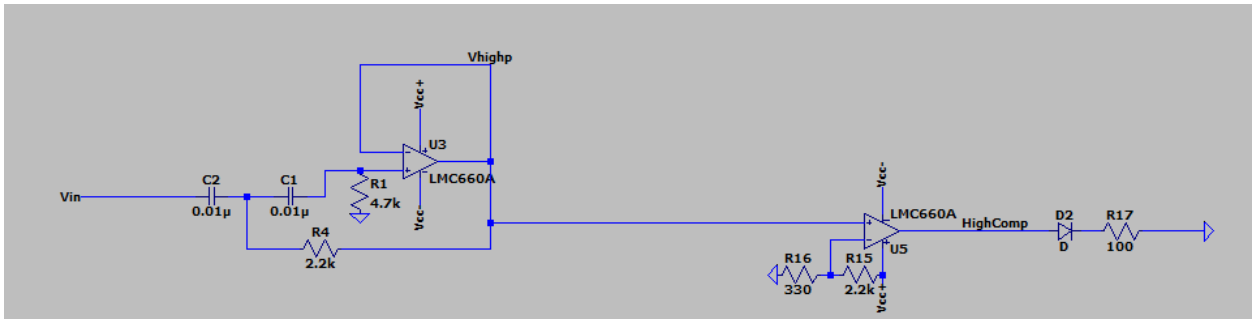
We attempted to implement a passive peak detector several times in LTspice, but it had no effect on our physical circuit. We believe that an active peak detector might have been more successful, but that would require 9 op amps in our system, which we didn't have enough chips to support. Even without the active peak detector, we believe our circuit still meets the outlined design requirements. The peak detector design below was attempted between the filters and the comparator.



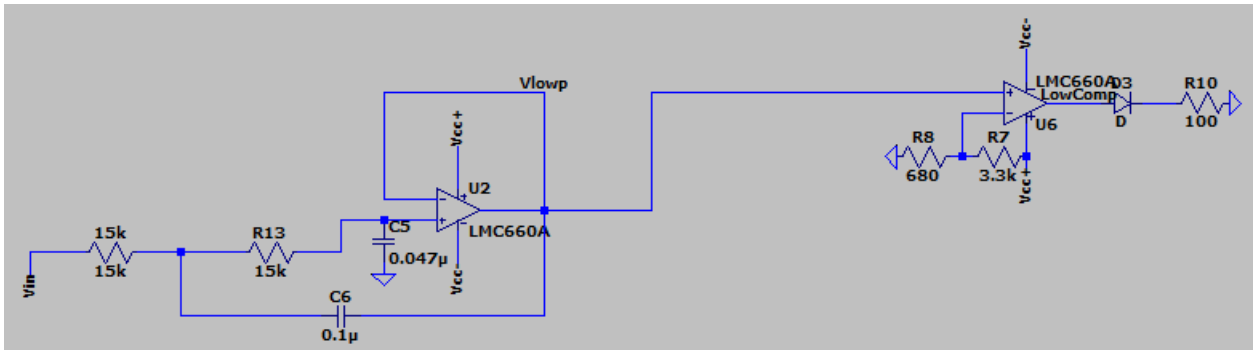
Finally, in our design, we used a comparator to take the opamp's output and feed it into Vin with an LED and a resistor across Vout. The hardest part was ensuring our Vref was at the correct voltage, so in our physical circuit, we experimented with many different resistor values to achieve the desired Vout at the right frequencies.



- **A circuit schematic that is clear and easily readable**



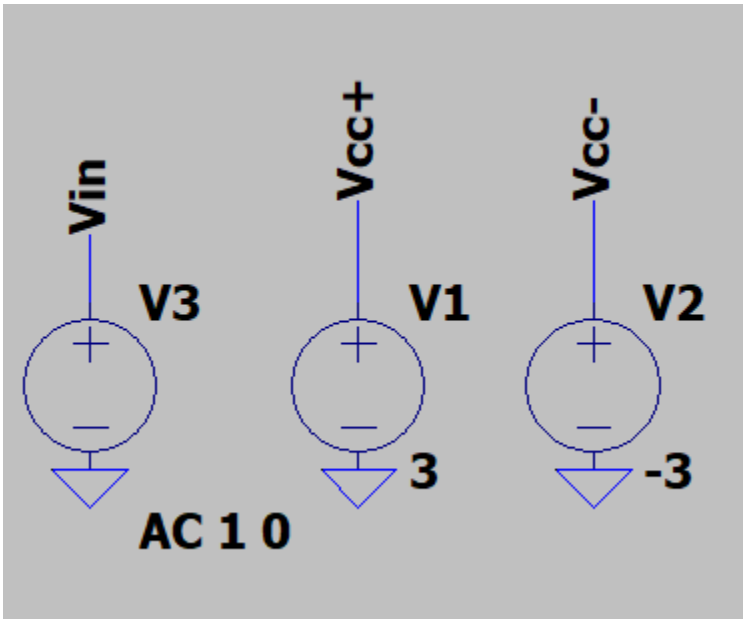
High Pass Filter



Low Pass Filter

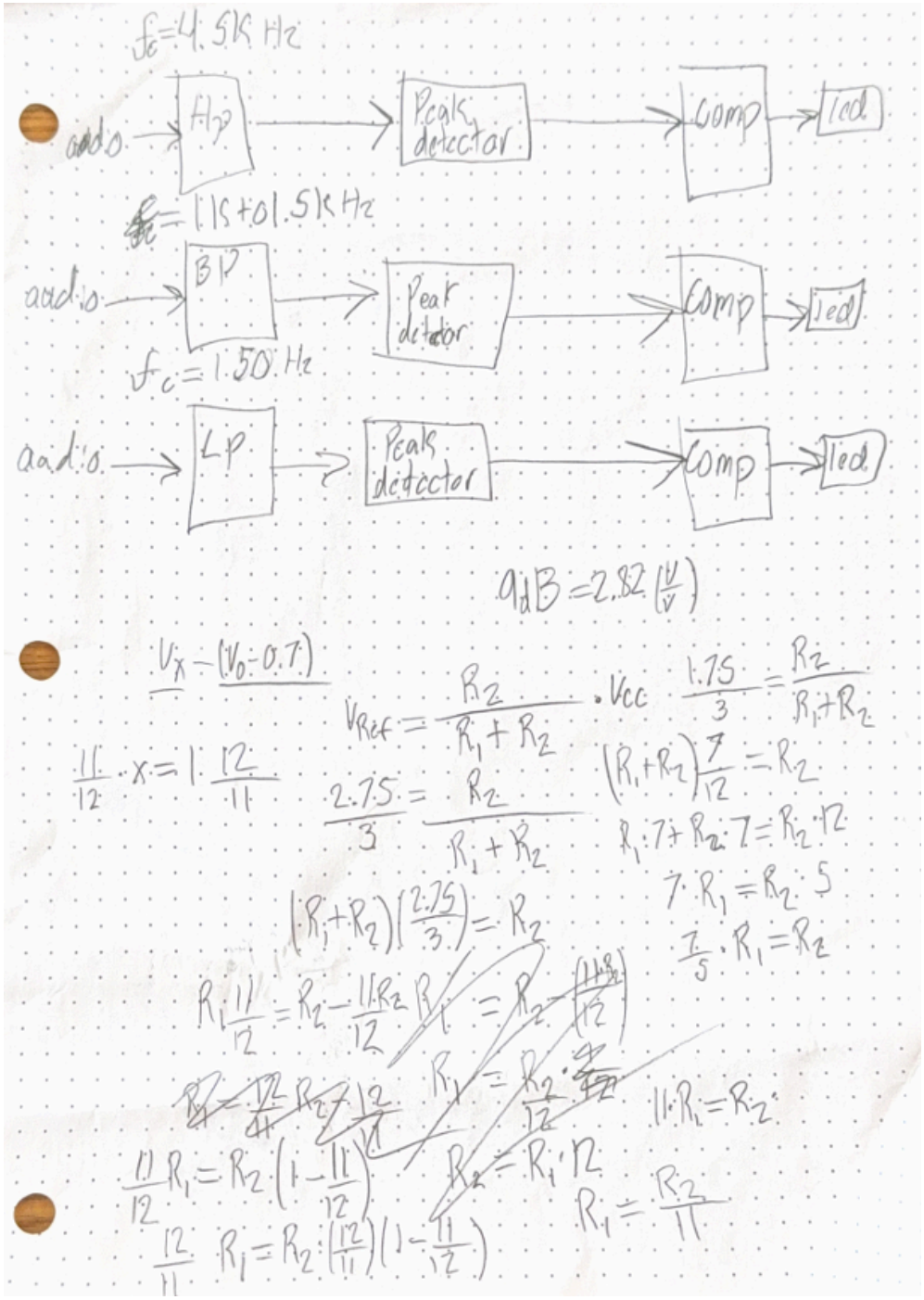


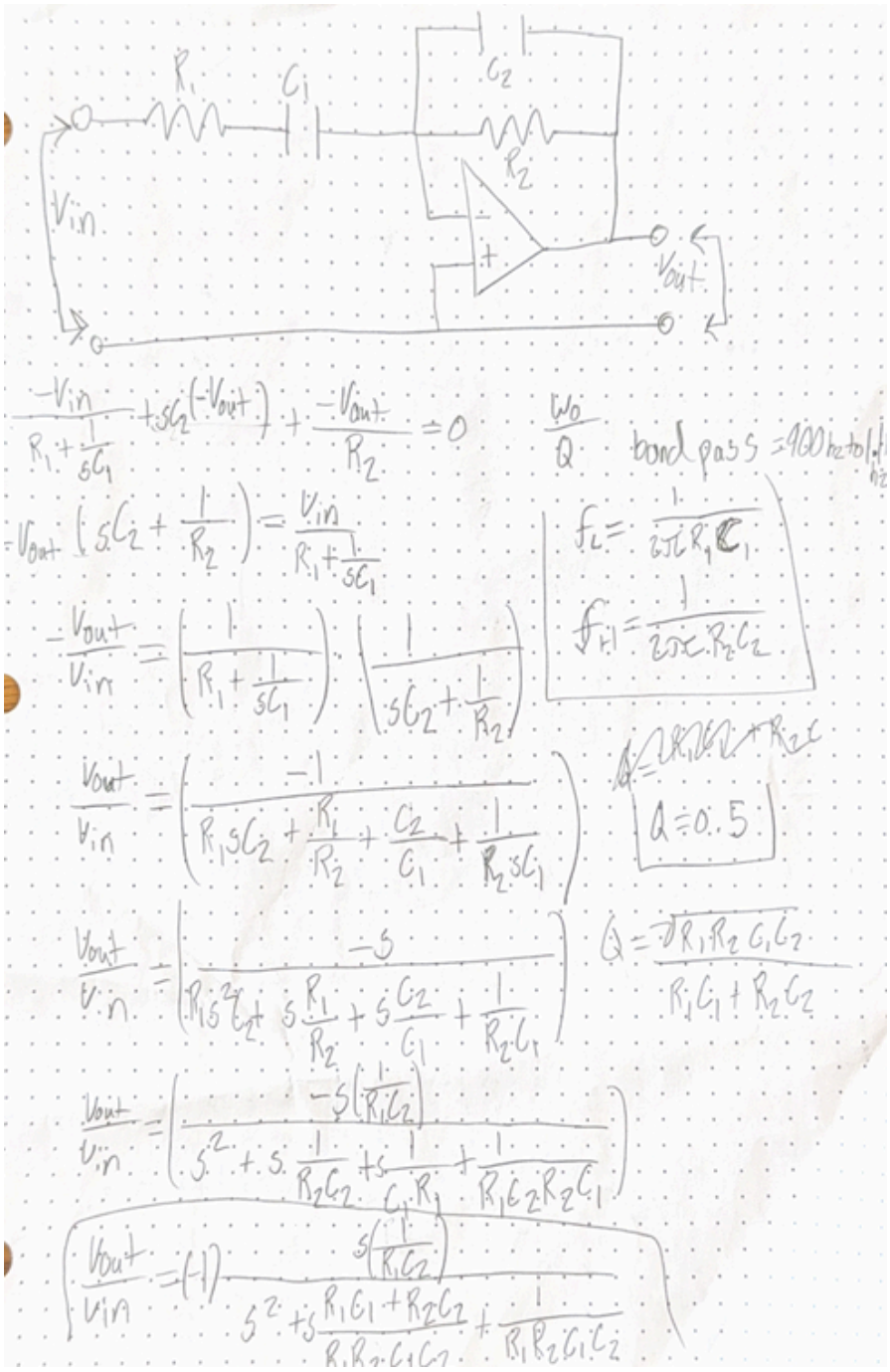
Band Pass Filter



Voltage Inputs (Vin is the L and R audio channel input)

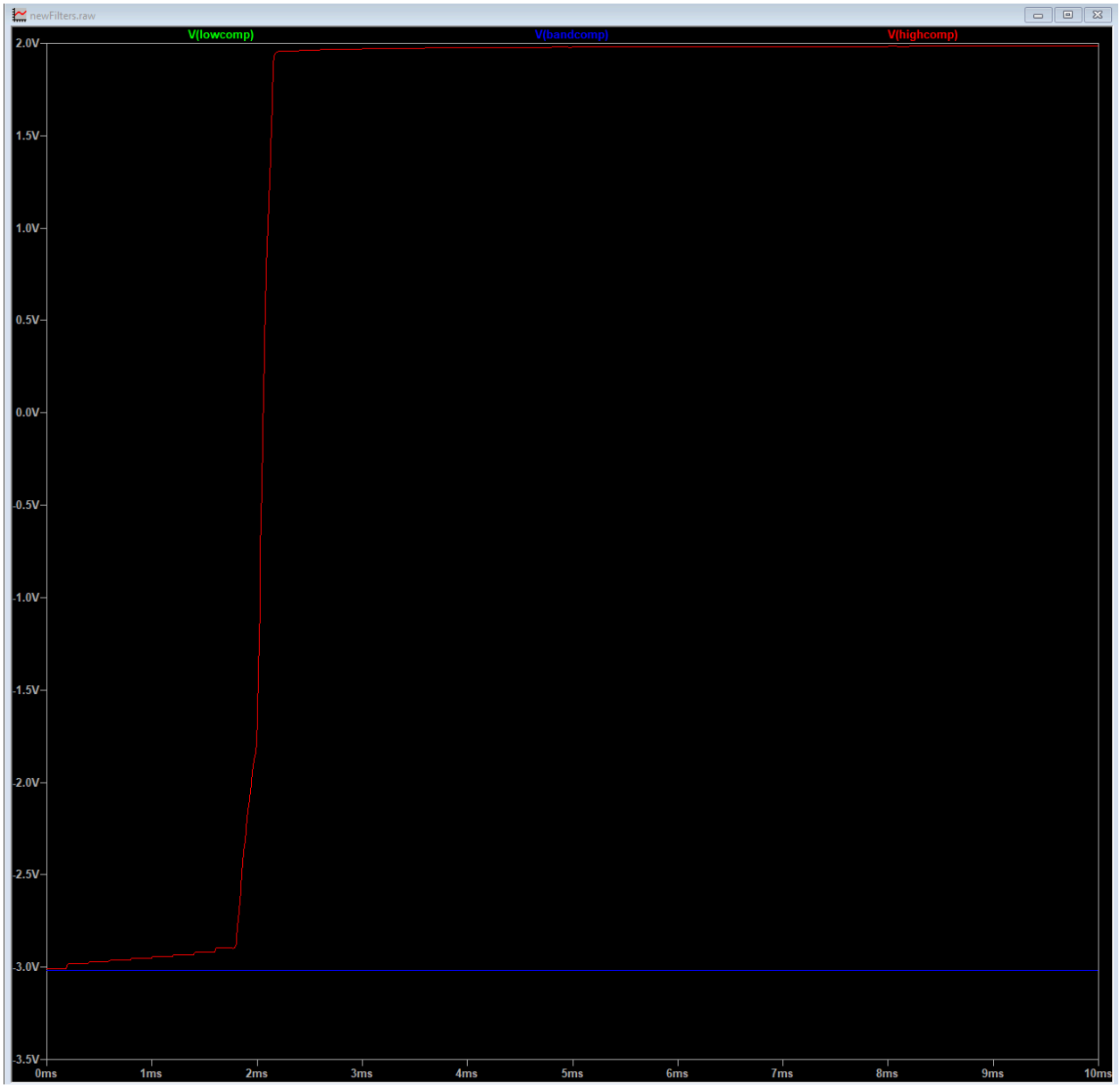
- **Design equations/analysis**



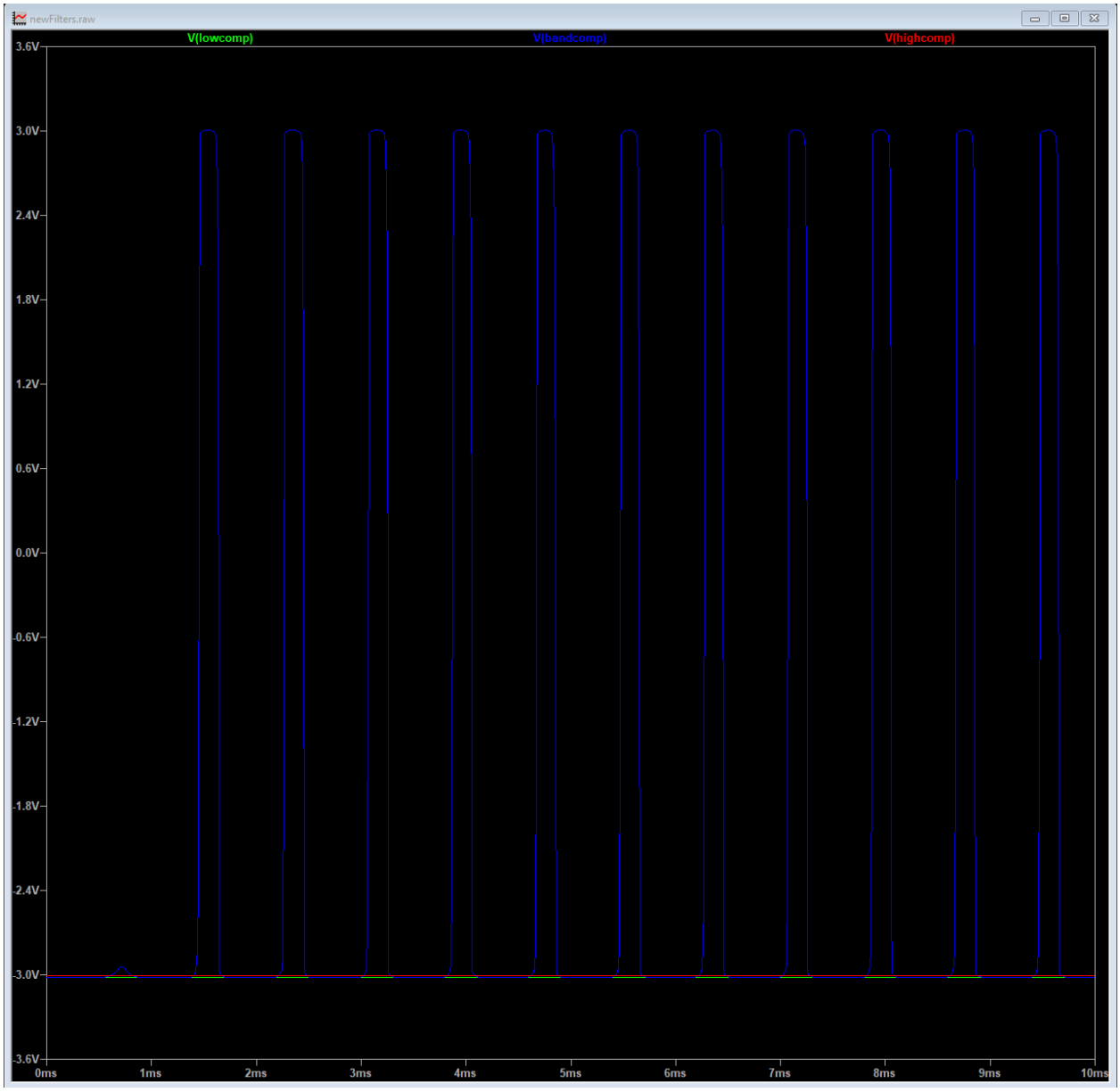


- **Simulation results**

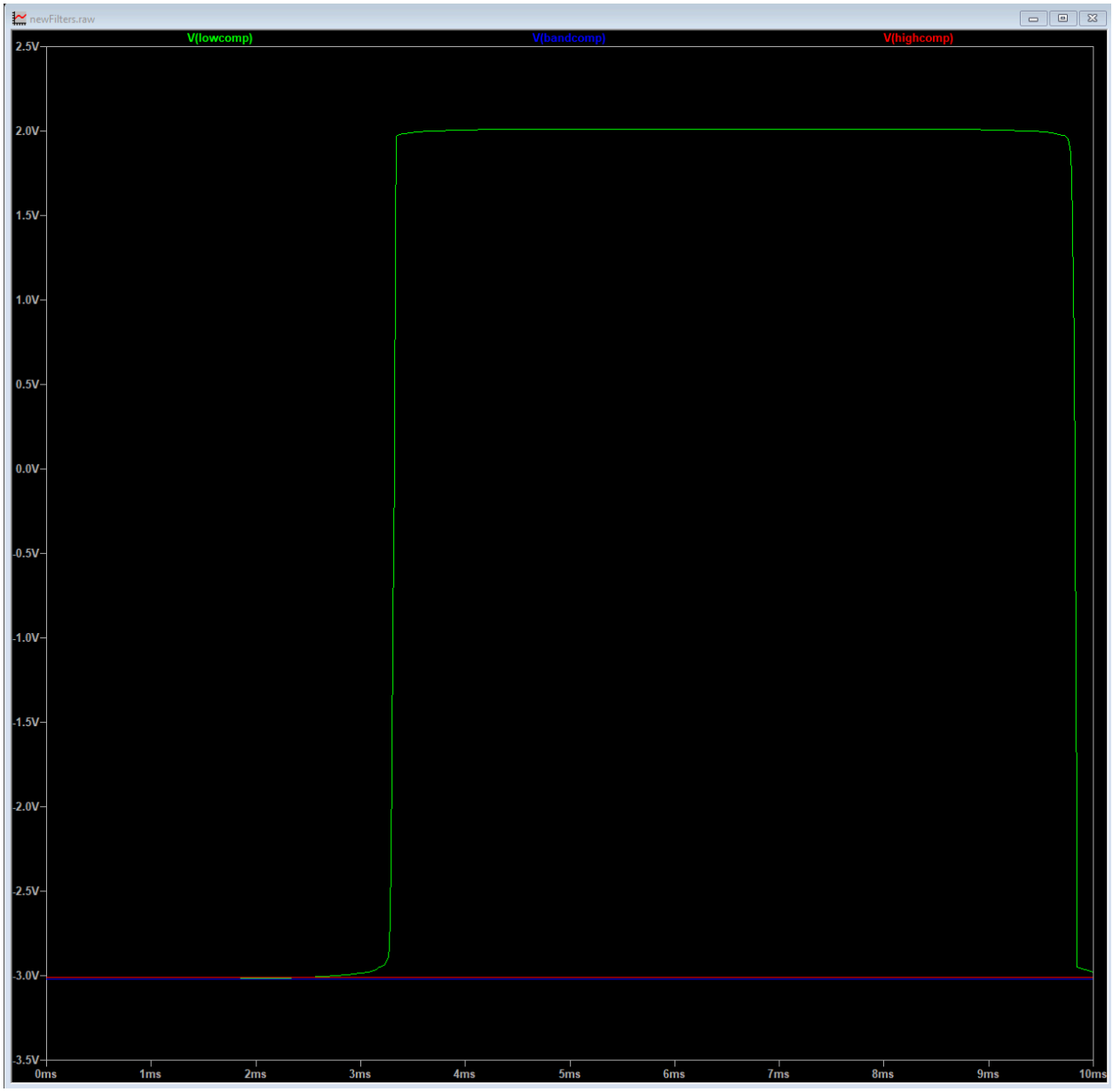
All of our circuits were attempted and redesigned in LTspice before trying anything in person. After that, the circuit's resistor values were adjusted in person to match the design requirements.



Comparator Outsputs with Sine(1V, 5000Hz)

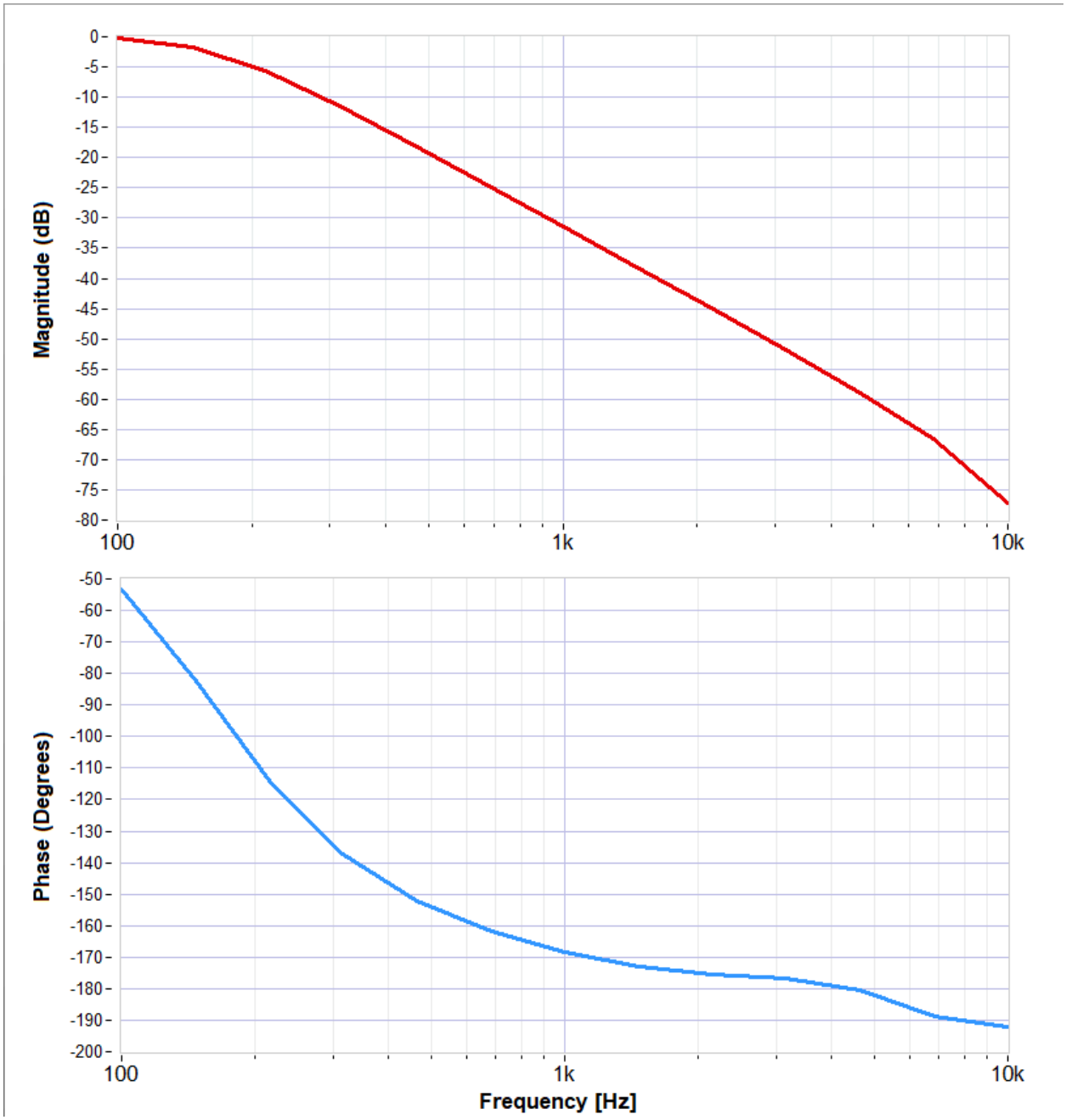


Comparator Outsputs with Sine(1V, 1000Hz)

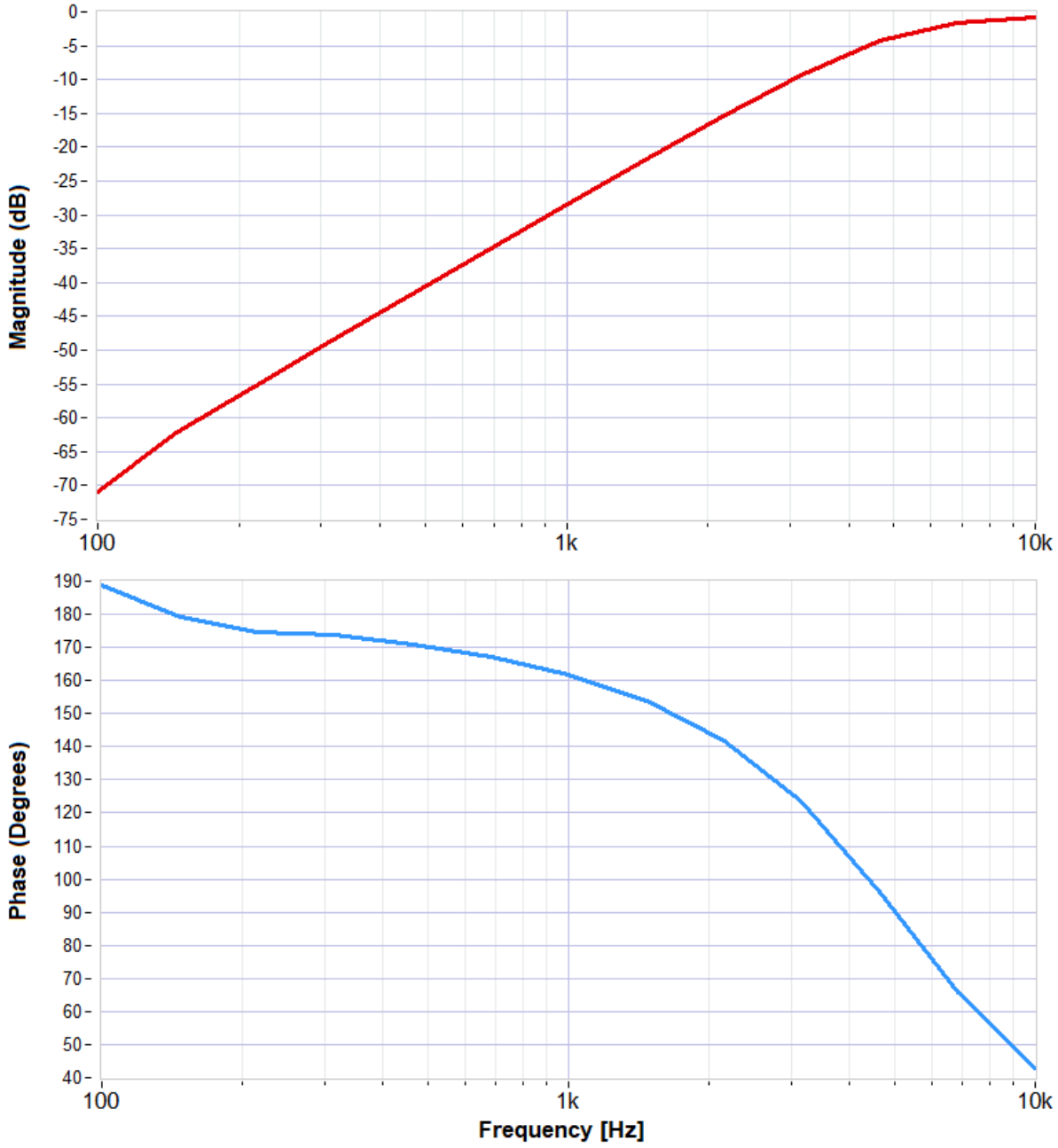


Comparator Outsputs with Sine(1V, 50Hz)

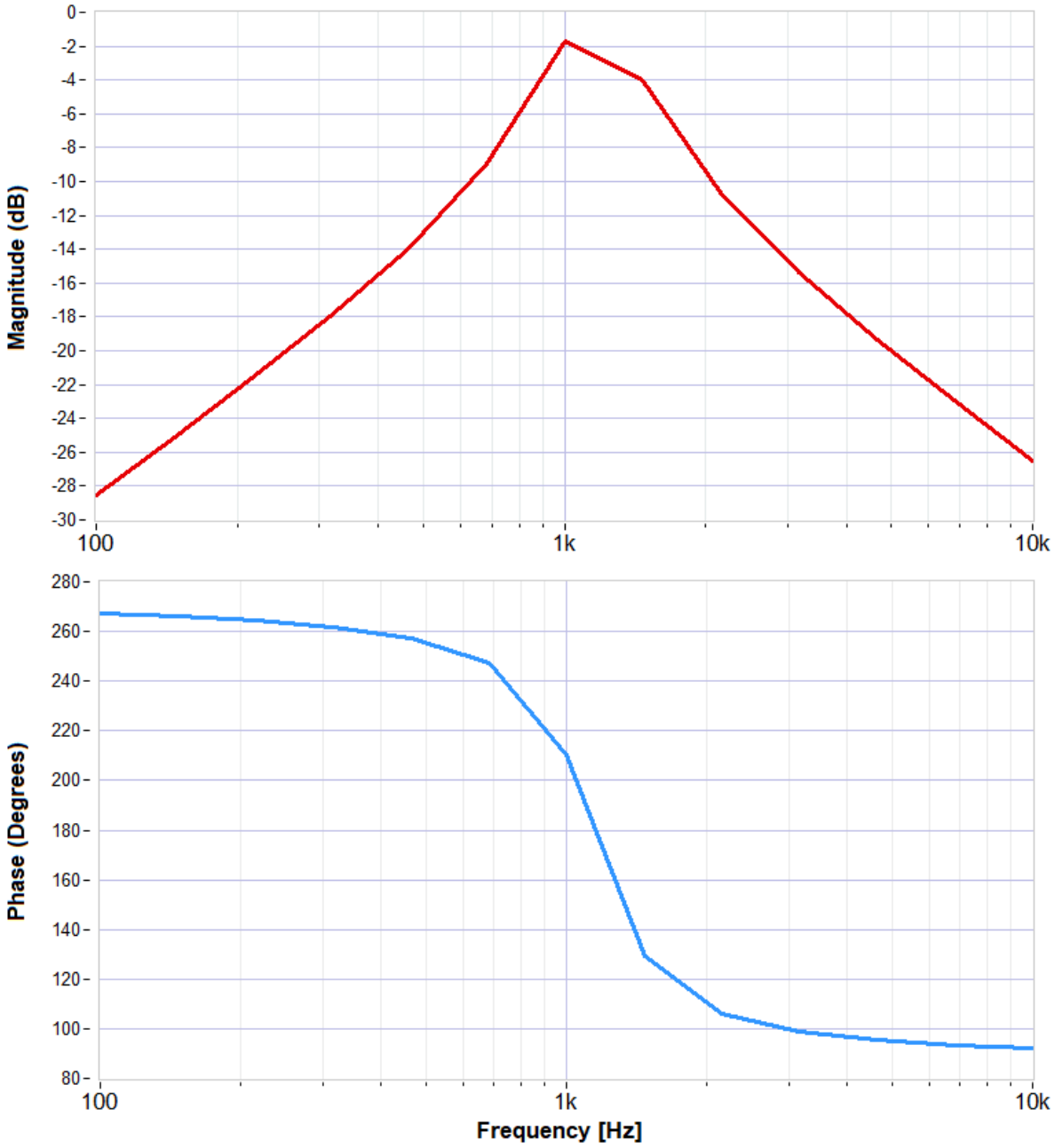
- **Any required measurement results listed in the project description**



Frequency Response Low Pass

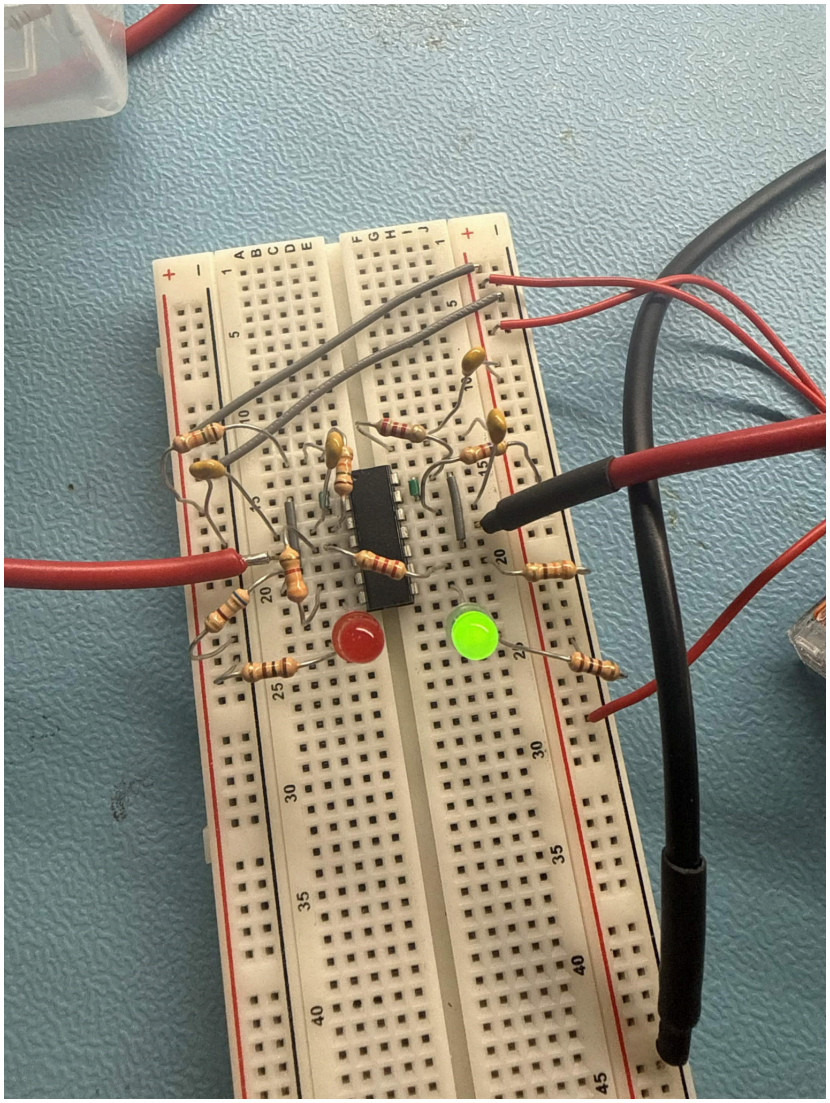


Frequency Response High Pass

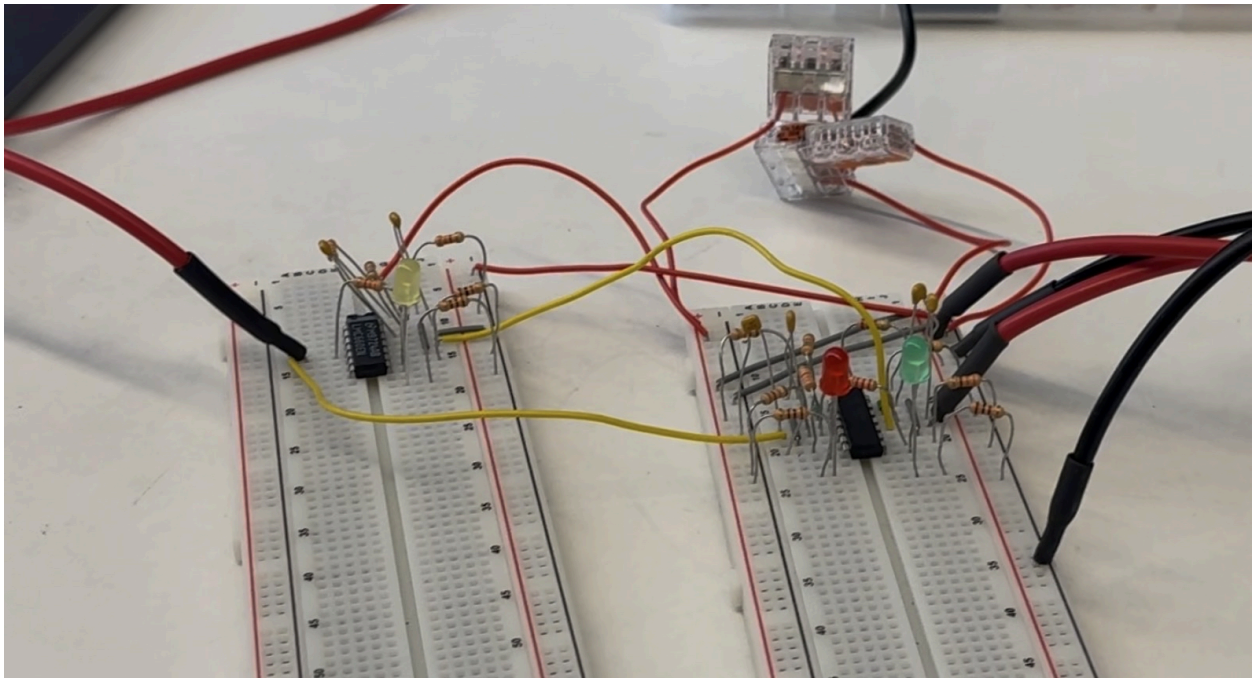


Frequency Response Band Pass

- **Photograph of your circuit**



Low and high pass filter



Band pass filter (left) and High and Low pass filter (right)

Troubleshooting and Analysis of Design Challenges

Due to time constraints, we were unable to demo during our lab sections for the live demonstration credit. This was due to a variety of reasons, but most notably was Michael's absence due to illness during the first lab session of the project, faulty op-amps, and an overemphasis on perfecting the equations and LTSpice schematics. During an out of lab session where we worked on our circuit, we noticed that we would obtain a very unusual output for our filters. After double checking our equations, testing in LTSpice, and fully rewiring the circuit, we still would get an output of about 4mA at the target cutoff frequency. This was incredibly frustrating, as we had the circuit constructed correctly, but did not understand why it was not functioning as expected. Eventually we swapped the op-amps and started getting relevant output voltages and determined that the op amp was faulty, in addition to testing it in the opamp tester in the lab. Lastly, in the demo lab section, while we had a fairly functional LTSpice schematic, we attempted to perfect it to have less error and have a smaller range of filtered frequencies with the equations in the slides. While this was beneficial to our project, it consumed a significant amount of time, as we didn't start building the circuit until about the last half hour of the lab. While we had all of the correct component values and a functional schematic, we simply didn't have enough time to construct the circuit which caused us to miss our original demo.

Conclusion

Overall, this project was a lot of fun, but we definitely encountered our fair share of problems along the way. We switched projects twice due to design challenges and time constraints, so this was our third project. We've learned how to analyze circuits well in this course, but designing circuits definitely required a different way of thinking. We struggled a lot with getting consistent outputs. One hindrance that we had at some point was that our op-amp had burned out, but we didn't realize it, and after working so long on the project, you don't realize when you are making simple mistakes, so there were times we built the circuit and weren't getting the correct output because we needed to adjust our resistor values.

This also plays into another thing we noticed. Beforehand, we had built everything in LTSpice, but, as I said, trying to perfect everything there wasted a lot of time compared to just troubleshooting the circuit in person. I think it was a good learning experience that the simulation can't consider all possibilities, because there was a circuit I accidentally wired incorrectly that gave us the output we wanted, but didn't have any output in LTspice. So we concluded that LTSpice is best as a reference, not a guide.

Bibliography

Components101. “Peak Detectors - How Does It Work and How to Build One Using Op-Amp?” *Components101*, 19 Feb. 2024, components101.com/articles/peak-detector-circuit-type-working-explanation-using-opamp.

Electronics Tutorials. “Op-Amp Comparator and the Op-Amp Comparator Circuit.” *Basic Electronics Tutorials*, 17 Sept. 2018, www.electronics-tutorials.ws/opamp/op-amp-comparator.html.

Module 3 – Lecture 4 (Second-Order Filters)

Nathan M. Neihart, “Module 3 – Lecture 4: Second-Order Filters,” EE 2300: Electronic Circuits & Systems, Iowa State University.

Module 3 – Lecture 5 (Filter Implementations)

Nathan M. Neihart, “Module 3 – Lecture 5: Filter Implementations,” EE 2300: Electronic Circuits & Systems, Iowa State University.

Module 3 – Lecture 6 (Filter Implementations)

Nathan M. Neihart, “Module 3 – Lecture 6: Filter Implementations,” EE 2300: Electronic Circuits & Systems, Iowa State University.